CMPG 322 - Decision Support Systems (DSS) 2
SU1 - Decision Theory 2



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Outline

1. Overview
2. Maximax - optimistic
3. Maximin - pessimistic
4. Criterion of realism (Hurwicz)
5. Criterion of realism (Hurwicz)
6. Equally likely (Laplace)
7. Minimax regret
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Criteria for decision making under uncertainty:

1. Maximax (optimistic)
2. Maximin (pessimistic)
3. Criterion of realism (Hurwicz)
4. Equally likely (Laplace)
5. Minimax regret

Maximax - optimistic

Used to find the alternative that maximizes the maximum payoff – maximax criterion:

- Locate the maximum payoff for each alternative;
- Select the alternative with the maximum number.

Maximax - optimistic

TABLE 3.2 Thompson's Maximax Decision

	STATE OF NATURE		
ALTERNATIVE	FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)	MAXIMUM IN A ROW ($)
Construct a large plant	200,000	−180,000	200,000 Maximax
Construct a small plant	100,000	−20,000	100,000
Do nothing	0	0	0

Maximin - pessimistic

Used to find the alternative that maximizes the minimum payoff – maximin criterion:

- Locate the minimum payoff for each alternative;
- Select the alternative with the maximum number.

Maximin - pessimistic

TABLE 3.3 Thompson's Maximin Decision

	STATE OF NATURE		
ALTERNATIVE	FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)	MAXIMUM IN A ROW ($)
Construct a large plant	200,000	−180,000	−180,000
Construct a small plant	100,000	−20,000	−20,000
Do nothing	0	0	0 Maximin

Criterion of realism (Hurwicz)

Often called weighted average, finds a compromise between optimism and pessimism:

- Select a coefficient of realism α, with $0 \leq \alpha \geq 1$
  - $\alpha = 1$ is perfectly optimistic
  - $\alpha = 0$ is perfectly pessimistic
- Compute the weighted averages for each alternative;
- Select the alternative with the highest value.

Weighted average $= \alpha$(best in row) + $(1-\alpha)$(worst in row)

Criterion of realism (Hurwicz)

For the large plant alternative use $\alpha = 0.8$ and for the small plant alternative use $\alpha = 0.8$:

TABLE 3.4 Thompson's Criterion of Realism Decision

	STATE OF NATURE		
ALTERNATIVE	FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)	CRITERION OF REALISM OR WEIGHTED AVERAGE (α = 0.8) ($)
Construct a large plant	200,000	−180,000	124,000 Realism
Construct a small plant	100,000	−20,000	76,000
Do nothing	0	0	0

Equally likely (Laplace)

Considers all the payoffs for each alternative:

- Find the average payoff for each alternative;
- Select the alternative with the highest average.

TABLE 3.5 Thompson's Equally Likely Decision

	STATE OF NATURE		
ALTERNATIVE	FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)	Blank ROW AVERAGE ($)
Construct a large plant	200,000	−180,000	10,000
Construct a small plant	100,000	−20,000	40,000
Do nothing	0	0	0

Equally likely

Minimax regret

Based on opportunity loss or regret, calculated as the difference between the optimal profit and actual payoff for a decision.

1. Create an opportunity loss table by determining the opportunity loss from not choosing the best alternative;
2. Calculate opportunity loss by subtracting each payoff in the column from the best payoff in the column;
3. Find the maximum opportunity loss for each alternative and pick the alternative with the minimum number.

TABLE 3.6 Determining Opportunity Losses for Thompson Lumber

STATE OF NATURE	
FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)
200,000 – 200,000	0 – (–180,000)
200,000 – 100,000	0 – (–20,000)
200,000 – 0	0 – 0

TABLE 3.7 Opportunity Loss Table for Thompson Lumber

	STATE OF NATURE	
ALTERNATIVE	FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)
Construct a large plant	0	180,000
Construct a small plant	100,000	20,000
Do nothing	200,000	0

Minimax regret

TABLE 3.8 Thompson's Minimax Decision Using Opportunity Loss

	STATE OF NATURE		
ALTERNATIVE	FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)	MAXIMUM IN A ROW ($)
Construct a large plant	0	180,000	180,000
Construct a small plant	100,000	20,000	100,000 ← Minimax
Do nothing	200,000	0	200,000

Summarising

- Maximax (optimistic)
- Maximin (pessimistic)
- Criterion of realism (Hurwicz)
- Equally likely (Laplace)
- Minimax regret

Questions

